

ALLEN Test - Mathematics

1. (3) Only (i), (iii), and (iv)
2. (3) Consistent with unique solution
3. (4) 6
4. (2) $x^2 - 2x + 1$
5. (2) 28th term
6. (2) -1, 7
7. (3) (2, -1)
8. (2) $\frac{BE}{EC}$
9. (4) P_2S_2
10. (4) $9\pi \text{ cm}^2$
11. (2) $\tan 30^\circ$
12. (4) $5\frac{1}{3}$
13. (3) $45(\sqrt{3}-1) \text{ m}$
14. (2) 40°

15. (4) 50cm

16. (1) $\frac{9}{13}$

17. (3) $\frac{1}{12}$

18. (2) ₹ 200,000

19. (2) Both A and R are true but R is not the correct explanation of A.

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21. Assuming $5 + 2\sqrt{3}$ is rational
i.e., $5 + 2\sqrt{3} = \frac{p}{q}$, $q \neq 0$ ~~$\& \text{HCF}(p, q) = 1$~~

$$2\sqrt{3} = \frac{p}{q} - 5$$

$$\sqrt{3} = \frac{p - 5q}{2q}$$

But it is given that $\sqrt{3}$ is irrational
 $\therefore \sqrt{3} \neq \frac{p - 5q}{2q}$

$\therefore 5 + 2\sqrt{3}$ is irrational

22. Let $\angle POQ = \angle ROS \Rightarrow x$

OR = OS

$\therefore \angle ORS = \angle OSR$

(vertically opposite angles)

(radii)

(angles opposite equal sides are equal)

In $\triangle OSR$

$$\angle ORS + \angle OSR + \angle ROS = 180^\circ$$

(angle sum property)

$$2\angle OSR = 180^\circ - x$$

$$\angle OSR = \frac{180^\circ - x}{2}$$

similarly, in $\triangle OPQ$, ($\because OP = OQ$)

$$\angle OPQ = \frac{180^\circ - x}{2}$$

$$\therefore \angle OSR = \angle ORS = \angle OPQ = \angle OQP$$

In $\triangle OSR$ & $\triangle OPQ$,

$$\angle ROS = \angle POQ$$

(vertically opposite angles)

$$\angle OSR = \angle OPQ$$

(proved above)

$$\therefore \triangle OSR \sim \triangle OPQ$$

(AA similarity)

23. ~~2m~~ Since QR is a tangent to R , $\angle PQR = 90^\circ$

$\therefore$ a radius is perpendicular to a tangent at point of contact.

Infinite number of such circles can be drawn.

$$24. \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta} = \frac{1 - \sqrt{3}}{1 + \sqrt{3}}$$

$$\frac{\frac{\cos \theta - \sin \theta}{\cos \theta}}{\frac{\cos \theta + \sin \theta}{\cos \theta}} = \frac{1 - \sqrt{3}}{1 + \sqrt{3}}$$

$$\frac{1 - \tan \theta}{1 + \tan \theta} = \frac{1 - \sqrt{3}}{1 + \sqrt{3}}$$

$$\therefore \tan \theta = \sqrt{3}$$

$$\therefore \theta = 60^\circ$$

25. $C_1 = 6 \text{ cm (radius)}$

~~Ques~~ $C_2 = 12 \text{ cm}$

$C_3 = 15 \text{ cm}$

$C_4 = 21 \text{ cm}$

A) Area of circle $C_1 = \frac{22}{7} \times 6^2$

$= 113.14 \text{ cm}^2$

Area of circle $C_4 = \frac{22}{7} \times 21^2$

$= 1386 \text{ cm}^2$

113.14 times 3.5 is not 1386, so Jaya is incorrect.

26. LCM of 6^4 , 8^2 and $k = 12^4$, find smallest value of k .

$6^4 = 2^4 \times 3^4$

$8^2 = 2^3 \times 2^3 = 2^6$

$\text{LCM}(6^4, 8^2, k) \Rightarrow 2^6 \times 3^4 \times k = 12^4$

$2^6 \times 3^4 \times k = 12^4$

$2^2 \times 2^4 \times 3^4 \times k = 12^4$

$2^2 \times k = \frac{12^4}{2^4 \times 3^4} = \frac{12^4}{6^4}$

$2^2 \times k = 2^4$

$k = \frac{2^4}{2^2}$

$\therefore k = 4$

27. $x^2 - 5x + 3(k-1) = 0$, $\alpha - \beta = 1$ - 0

$\alpha + \beta = 5$ - (2)

From (1) + (2),

$\alpha + \beta = 5$

+ $(\alpha - \beta = 1)$
 $\hline 2\alpha = 6$

$$\therefore \alpha = 3$$

$$\therefore \beta = 2$$

$$\alpha\beta \Rightarrow \sigma = 3(k-1)$$

$$2 = k-1$$

$$\therefore k = 3$$

28(B) ~~28(B)~~ Let the fraction be $\frac{x}{y}$

$$\frac{x-2}{y} \times \frac{1}{3}$$

$$3(x-2) = y$$

$$3x - 6 = y$$

—— (1)

$$\frac{x}{y-1} \times \frac{1}{2}$$

$$2x = y - 1$$

$$2x = 3x - 6 - 1$$

$$2x = 3x - 7$$

$$\therefore x = 7$$

$$\therefore y = 15$$

$$\therefore \frac{x}{y} = \frac{7}{15}$$

$$29. (B) \text{ Area of playground } \Rightarrow 22176 \text{ cm}^2 = \pi r^2$$

$$\frac{22176 \times 7}{22} = r^2$$

$$7056 = r^2$$

$$\therefore r = 84 \text{ cm} = 0.84 \text{ m}$$

$$\text{Cost of fencing playground} = ₹ 50 \times 2\pi r$$

$$= 50 \times 2 \times \frac{22}{7} \times 0.84$$

$$= ₹ 264$$

$$30. \frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \tan \theta + \cot \theta \quad (\text{To prove})$$

$$\text{LHS: } \frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = \frac{\tan \theta}{1 - \frac{1}{\tan \theta}} + \frac{\frac{1}{\tan \theta}}{1 - \tan \theta}$$

$$= \frac{\tan^2 \theta}{\tan \theta - 1} - \frac{1}{(\tan \theta - 1) \tan \theta} = \frac{\tan^3 \theta + 1}{(\tan \theta - 1) \tan \theta}$$

$$= \frac{(\tan \theta - 1)(\tan^2 \theta + \tan \theta + 1)}{(\tan \theta - 1) \tan \theta} = \frac{\tan^2 \theta + \tan \theta + 1}{\tan \theta}$$

$$= \frac{\tan^2 \theta}{\tan \theta} + \frac{\tan \theta}{\tan \theta} + \frac{1}{\tan \theta} = \tan \theta + 1 + \cot \theta$$

$\therefore$ Proved

31. Median = 16

$\Sigma f = 70$

Class	0-5	5-10	10-15	15-20	20-25	25-30	30-35	35-40
f	12	a	12	15	b	6	6	4
(f	12	12+a	24+a	39+a	39+a+b	45+a+b	51+a+b	55+a+b

$70 = 55 + a + b$

$a + b = 15$

$\frac{n}{2} = 35$, median = 16

median class = 15-20, $l = 15$

$f = 15$, $(f = (24+a))$, $h = 5$

median = $l + \left(\frac{\frac{n}{2} - cf}{f} \right) \times h$

$16 = 15 + \frac{35 - 24 - a}{15} \times 5$

$1 = \frac{11 - a}{3}$

$3 = 11 - a$

$\therefore a = 8$

$\therefore b = 15 - 8 \Rightarrow 7$

32. (A) Let the greater number be x , and smaller number be y

$x^2 - y^2 = 180$, $y^2 = 8x$

$x^2 - 8x = 180$

$x^2 - 8x - 180 = 0 \rightarrow x^2 - 18x + 10x - 180 = 0$

$x(x - 18) + 10(x - 18) = 0$

$(x + 10)(x - 18) = 0$

$\therefore x = 18$

($\because x$ cannot be negative
 $x \neq -10$)

$$y^2 = 8x = 8 \times 18 = 144$$

$$y = \sqrt{144}$$

$$\therefore y = 12$$

33. $OD = DX$, $BD = CD$

$\therefore BC$ and OX bisect each other

$\therefore OBXC$ is a parallelogram

~~So~~, $BE \parallel XC$ and $BX \parallel CF$

In $\triangle ABX$, $\frac{AF}{FB} = \frac{AO}{OX}$ — (1) (by BPT)

In $\triangle AXC$, $\frac{AE}{EC} = \frac{AO}{OX}$ — (2) (by BPT)

From (1) & (2),

$$\frac{AF}{FB} = \frac{AE}{EC}$$

$\therefore EF \parallel BC$ — (i) (by converse BPT)

$$\frac{OX}{AO} = \frac{FB}{AF}$$

$$\frac{OX}{AO} + 1 = \frac{FB}{AF} + 1 \Rightarrow \frac{OX + AO}{AO} = \frac{FB + AF}{AF}$$

$$\frac{AX}{AO} = \frac{AB}{AF}$$

$\therefore OA : AX = AF : AB$ — (ii)

34(B) $l = 8m 50cm = 850cm$

$b = 6m 25cm = 625cm$

$h = 4m 75cm = 475cm$

longest rod that measures the room dimensions
 $= HCF(850, 625, 475)$

$$850 = 2 \times \underline{5} \times 5 \times 17$$

$$625 = \underline{5} \times 5 \times \underline{5} \times 5$$

$$475 = \underline{5} \times 5 \times 19$$

$$\text{HCF}(850, 625, 475) = 5^2 = 25 \text{ cm}$$

Measure of longest rod = 25 cm

35. $a = 3$, $l = 83$, $S_n = 903$

$$S_n = \frac{n}{2} (a + l)$$

$$903 = \frac{n}{2} (3 + 83) \rightarrow 903 = \frac{n \times 86}{2}$$

$$\therefore n = \frac{903}{43} = 21$$

$$l = a_n = a + (n-1)d$$

$$83 = 3 + (21-1)d$$

$$80 = 20d$$

$$\therefore d = 4$$

36. S_n
 ~~S_n~~ = 118000

$$a = 1000$$

$$d = 100$$

$$(i) a_{30} = a + 29d = 1000 + 29(100) \\ = 1000 + 2900$$

$$\therefore a_{30} = 3900$$

$$(iii) S_n = \frac{n}{2} (a + l)$$

$$118000 = \frac{40}{2} (1000 + l)$$

$$5900 = 1000 + l$$

$$\therefore l = 4900$$

$$(iii) S_{30} = \frac{30}{2} (2a + (30-1)d)$$

$$= 15 (2000 + 2900)$$

$$= 15 \times 4900$$

$$\therefore S_{30} = 73500$$

37. $AP = 31, 27, 23, \dots, 3$

(i) $a = 31$

$$d = 27 - 31 = (-4)$$

$$a_n = a + (n-1)d$$

$$= 31 + (n-1)(-4)$$

$$= 31 - 4n + 4$$

$$\therefore a_n = 35 - 4n$$

(ii) $a_5 = 35 - 4(5) = 35 - 20$

$$\therefore a_5 = 15$$

(iii) $11 = 35 - 4n$

$$-24 = -4n$$

$$\therefore n = 6$$

38. $AD = 5m$

$AB = 1.3m$

$\angle C = 60^\circ$

(i) $BD = AD - AB$
 $= 5 - 1.3$

$\therefore BD = 3.7m$

(ii) $BC = ?$

$\sin 60^\circ = \frac{BD}{BC}$

$\frac{\sqrt{3}}{2} = \frac{3.7}{BC}$

~~BC~~ $BC = \frac{2 \times 3.7}{\sqrt{3}} = \frac{3.7 \times 2 \times \sqrt{3}}{3}$

$\therefore BC = 2.46\sqrt{3}m$

(iii) $\tan 60^\circ = \frac{BD}{DC}$

$\sqrt{3} = \frac{5}{DC}$

$DC = \frac{5}{\sqrt{3}}m$

$\sqrt{3} = \frac{3.7}{DC}$

$DC = \frac{3.7}{\sqrt{3}}m$

$\therefore DC = \frac{3.7\sqrt{3}}{3}m$

